Using a Z-score Approach to Combine Value and Momentum in Tactical Asset Allocation

PENG WANG AND LARRY KOCHARD

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PENG WANG
is a quantitative investment analyst at the Georgetown University Investment Office in Washington, DC.
pw35@georgetown.edu

LARRY KOCHARD
is the chief executive officer at the University of Virginia Investment Management Company in Charlottesville, VA.
lkochard@virginia.edu

Research has shown that value and momentum deliver abnormal positive expected returns in a variety of markets and asset classes at the security level (Asness, Moskowitz, and Pedersen [2008]). The key issue addressed in this article is whether such an effect can be observed across asset classes at the index level in a tactical asset allocation framework. Asset allocation has been shown to be an important factor in portfolio performance attribution (Brinson, Singer, and Beebower [1986]). Ibbotson and Kaplan [2000] also point out that most of the variation in a typical fund's return comes from the market environment. In such ever-changing market environments, the overall portfolio performance can be significantly affected by tactical asset allocation (TAA) (Lee [2000]). Specifically, by tactically adjusting the relative weights of asset classes based on their perceived value and momentum attractiveness, we improve the risk-adjusted returns on a given strategic asset allocation. The strategic asset allocation here is a representative mix of seven diversified asset classes, including global equity, investment-grade bonds, high-yield bonds, cash, treasury inflation-protected securities (TIPS), commodities, and real estate. For practitioners, the model provides a straightforward dynamic top-down approach to tactical asset allocation in accordance with the ever-changing market environments (Li and Sullivan [2011]).

DATA AND METHODOLOGY

Exhibit 1 provides an overview of the seven asset classes included in the framework and the indices for the value and momentum signals used in this article. Each asset class is selected to provide a unique set of return and risk characteristics so that a portfolio of the asset classes provides opportunities for growth as well as protection against both deflation and inflation risks. Equity includes both U.S. and non-U.S. equity, as the distinction between the two has lost some of its meaning over time. Fixed income is comprised of investment grade (yield curve risk), high yield (credit risk) and cash. Real assets include commodities, real estate, and TIPS that protect investors during an inflationary regime.

The indices used to represent the seven asset classes are (Exhibit 1): the Morgan Stanley Capital International ACWI Index (MSCI ACWI), Barclays Capital Aggregate Bond Index (Barclays Agg.) gross return, Merrill Lynch High Yield Master II (MLHY II) total return, Merrill Lynch 91-Day Treasury (T-Bill), 10-year on-the-run TIPS, Goldman Sachs Commodity Index (GSCI) total return,

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EXHIBIT 1
Asset Classes and Indices

	Asset Class	Index
Equity	Global Equity	MSCI ACWI
	Investment Grade	Barclays Agg.
Fixed Income	High Yield	MLHY II
	Cash	T-Bill
	TIPS	10-Yr on-the-run TIPS
Real Estate	Real Estate	NAREIT
	Commodity	GSCI

and the National Association of Real Estate Investment Trusts Index (NAREIT) total return.

Momentum describes the persistence between an asset's return and its recent relative performance history. Positive momentum effects have been found in securities (Jegadeesh and Titman [1993]), international markets (Rouwenhorst [1998], Asness, Moskowitz, and Pedersen [2008]), sectors and industries (Moskowitz and Grinblatt [1999]) and asset classes (Blitz and Van Vliet [2008]). We follow the convention to define our momentum signals using past return data. For each asset class, the return over a simple moving average (SMA) of trailing twelve-month ending price[1] lagged by one month is used. We first test momentum strategy in asset classes on a relative basis—the relative winners (losers) will be given higher (lower) exposures, even if the winners could just have suffered smaller losses than others. Furthermore, we also investigate momentum strategy on an absolute basis; that is, we only increase allocation to the winners with positive returns, and reduce exposures to any asset classes with negative returns and increase the cash reserve correspondingly. This modification not only effectively preserves capital, but also explicitly reserves more cash dry powder for the value strategy to fire in the combined tactical asset allocation model. And the overall portfolio risk-adjusted performance is improved significantly.

On the contrary to momentum, it is less straightforward to construct a cross-asset class value strategy, because no obvious valuation measure is applicable to every asset class. The starting point of the approach is a simple yield measure for equity and fixed income (Blitz and Van Vliet [2008]). We use book-to-price (B/P) for equity assets, cash-flow-to-price for REITs, yield spread between BAA and 10-yr treasury for investment grade, and the standard yield-to-maturity for high yield, cash, and TIPS. For commodities, we develop a backwardation-contango strategy defined by (next month futures price—current month futures price)/next month futures price. If this signal is negative, the commodity is in contango; and if it is positive, it is in backwardation. Backwardation suggests a value situation because of the expected positive roll yield. All of the yield measurements share the same feature that a larger value implies a more attractive valuation. The data used are from January 1986[2] to December 2010, except TIPS, which is from March 1997 to December 2010. The yields on BAA, 10-yr treasury, T-Bill, and TIPS are from the Federal Reserve System website; cash flow/price for NAREIT is based on Goldman Sachs' respective U.S. REIT universe.[3] The backwardation/contango signal is calculated from GSCI generic futures prices from Bloomberg.

A big challenge to applying a value strategy across asset classes comes from the fact that not all value measures are directly comparable.[4] In order to account for the inherent structural differences across asset classes, while not introducing a meaningful bias, we standardize our simple yield approach with the Z-score under the expanding window approach to avoid look-ahead bias. The Z-score measures the number of standard deviations the signal is from its historical mean. It is calculated for each month t with one-month lag to ensure the availability of data; meaning using its entire historical data up to month $t-1$, based on the following formula in Equation (1).

$$Z_t = \frac{V_{t-1} - mean(V_{0,t-1})}{Std(V_{0,t-1})} \tag{1}$$

Exhibit 2 gives a snapshot of both the simple yield and the Z-score measurements for value at the end of 2010. As we can see, it is less meaningful to directly compare the basic yield measurement of equity (B/P) to the yield of investment grade, or the yield of investment grade to that of high yield. Meanwhile, the standardization process indeed scaled the valuation measurements to the same range so that a direct comparison of the valuation Z-scores across asset classes is more appropriate. In the same spirit as our modification to the momentum strategy, we also consider both relative and absolute valuations for each asset class. Our Z-score approach not only enables us to directly compare relative valuation across asset classes, but offers insight about each asset class's absolute valuation level by its own standard. We

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EXHIBIT 2
Basic Yields Z-Scores

1989.1– 2010.12	MSCI ACWI	Barclays Agg.	MLHY II	T-Bill	TIPS	GSCI	NAREIT
Basic Yields							
Mean	0.43	2.24	11.08	3.91	2.49	0.00	0.09
Median	0.42	1.99	10.50	4.43	2.24	0.00	0.08
Max.	0.80	6.01	21.71	9.14	4.33	0.08	0.13
Min.	0.25	1.29	7.43	0.03	0.53	–0.04	0.05
Z-Scores							
Mean	–0.05	0.42	–0.47	–0.83	–0.88	–0.38	0.51
Median	–0.17	0.05	–1.01	–0.68	–1.23	–0.48	0.51
Max.	4.93	6.48	4.41	2.96	2.90	4.32	4.47
Min.	–2.50	–2.20	–2.45	–2.47	–3.95	–2.68	–2.11

only identify any asset class in good valuation, when it is both absolutely cheap and relatively cheaper than others at the same time.

Exhibit 3 shows historical Z-scores under the expanding window approach for various asset classes. We can see that global equity was quite cheap[5] in the early 90s, became expensive and peaked around 1999 to 2000, and then kept dropping value during the 2000 to 2003 tech bubble burst. Equity was more than three standard deviations cheap at the bottom of the global financial crisis from November 2008 to April 2009. During the same period, REITs and credits also offered attractive valuations, while commodities and TIPS did not. In this sense, the Z-score approach helps identify market peaks and bottoms for each asset class.

THE TACTICAL ASSET ALLOCTION FRAKEWORK

We proceed by constructing two strategic allocation portfolios. We then use these portfolios as benchmarks and compare our TAA model in forms of performance. The first portfolio equally weighs all the asset classes. The other portfolio is more conventional with the high-equity concentration typically used by institutional investors. Since TIPS was not introduced until 1997, there are six asset classes before March 1998 and seven asset classes including TIPS after.[6] Exhibit 4 provides an overview of the two base allocations.

Momentum

At the end of every month, each asset class is ranked based on its respective momentum signal. The ranking is used to decide the tactical weights. For asset class i, the weight is given in Equation (2) (Asness, Moskowitz, and Pedersen [2008]).

$$W_m(\mathrm{i}) = Base(\mathrm{i}) + R * (rank(i) - average(rank)) \quad (2)$$

For the case of the seven asset classes, the average rank by definition is four. The adjustments made to the asset classes are always –3xR, –2xR, –R, 0, R, 2xR, and 3xR based on rankings from one to seven. Consequently, the total net adjustment is 0, and the summation of the weights remains the same as before. The adjusting basis R is a parameter that can be changed depending on the investor's risk preference. A higher value of R means higher risk tolerance to take short and leveraged positions.[7] For now, it is set at 2%, which results in small deviations from the benchmarks and satisfies the no-short constraint[8] most of the time.

So far, the momentum signal is compared relatively across asset classes. We further modify the momentum strategy by including absolute performances (Faber [2007]). The exposure to any asset class with a negative return is reduced to 0[9] and cash is increased correspondingly (Equation (3)). Cash as a source of tail risk protection is often underrated. As shown in the results, this simple modification increases risk-adjusted return (higher Sharpe ratio) significantly in the long run, because of capital preservation. More importantly, holding cash also allows us to invest when the opportunity set looks better. In the combined model, this modification tends to increase the investing power of the value strategy at the right time.

$$W_{mt}(cash) = W_m(cash) + \sum W_m(i) \quad (3)$$

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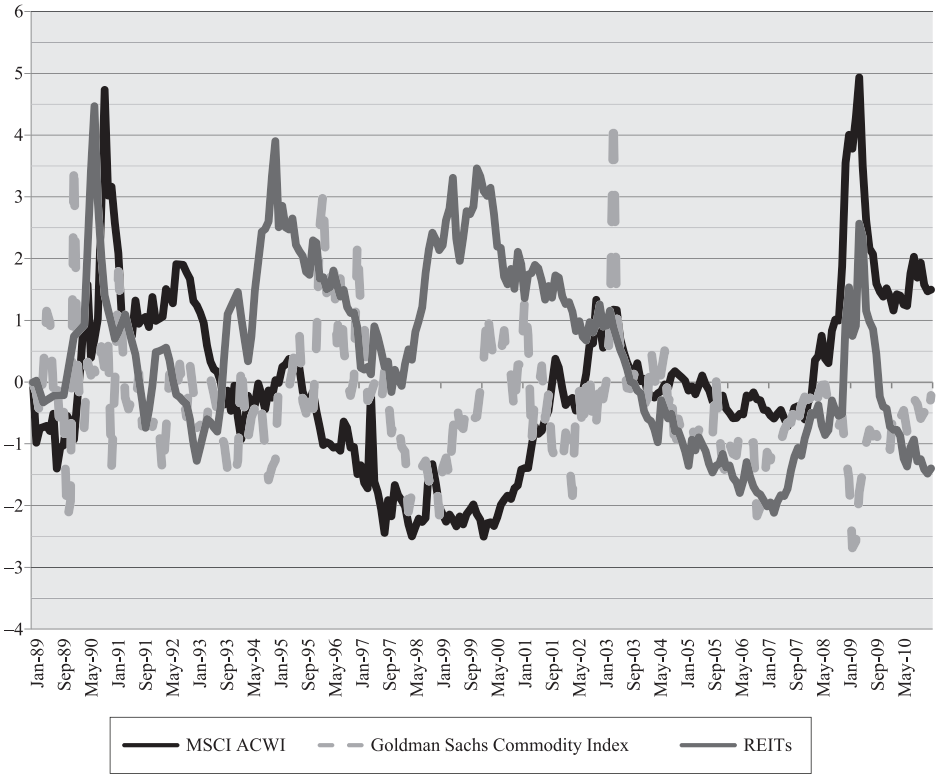
EXHIBIT 3
Historical Z-Scores Expanding Window



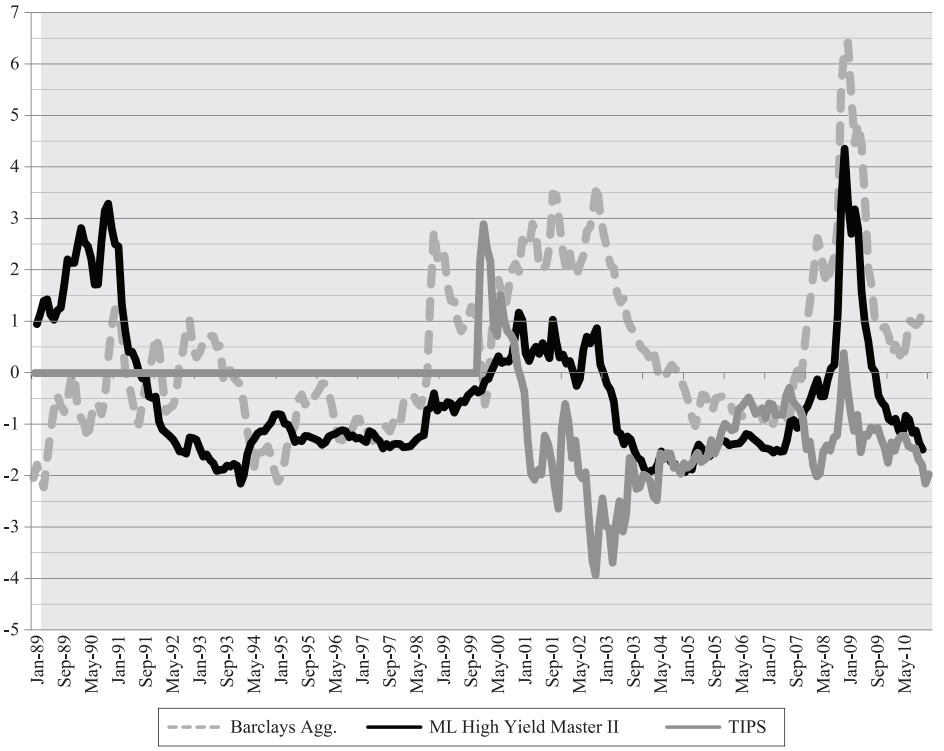


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EXHIBIT 4
Benchmark Weights

	Global Equity	Investment Grade	High Yield	Cash	TIPS	Commodity	Real Estate
Equal							
before 1998/03	1/6	1/6	1/6	1/6	0	1/6	1/6
after 1998/03	1/7	1/7	1/7	1/7	1/7	1/7	1/7
Hypothetical							
before 1998/03	45.67%	18.17%	11.17%	7.67%	0	8.67%	8.67%
after 1998/03	45.0%	17.5%	10.5%	7.0%	4.0%	8.0%	8.0%

$W_{mt}(i) = 0\%$, any asset class i with negative momentum.

Value

We calculate the valuation Z-scores, which are used to identify the under/over-valued asset classes each month. The asset class weights are adjusted from their rankings as with the momentum strategy (Equation (4)).

$$W_v(\mathrm{i}) = Base(i) + R * (rank(i) - average(rank)) \quad (4)$$

As an effort to avoid the value trap, we implement our value strategy in a nonlinear fashion with critical values. For example, if we use a critical value of 2, then the asset classes with valuation Z-scores between -2 and 2 will be treated as 0 and share the same rank in Equation (4). In other words, we first identify those asset classes whose valuation is at least 2 standard deviations away from their historical means (absolute valuation) and then compare them across asset classes (relative valuation). This nonlinear approach reflects the idea that the mean-reversion value strategy works better in extreme situations. Exhibit 5 gives us an overview of how the critical values affect the performance on the equally weighted base allocation. As the critical value decreases, the condition to be considered absolutely under/overvalued is loosened and the nonlinear value strategy is triggered more often. For the critical value 2, the value strategy is triggered in about 61.2% of the 264 months. Using a critical value of 1, every month we see at least 1 asset class either overvalued or undervalued (100% trigger ratio). For the critical value 0, the model ranks the raw Z-scores and applies a linear approach. The hit ratio is defined as the success rate of outperforming the benchmark when the value strategy is triggered. As the critical value decreases, so does the hit ratio; the value strategy makes more bets with lower success rates, which affirms the idea that the mean-reversion value strategy predicts better in extremes. At a critical value of 3, although the hit ratio increases, the annual excess return is lower because the value strategy is triggered less often. The 20-year Sharpe ratio, annual excess return, and information ratio are optimized around the critical value of 1.5, which offers a good balance between the number of bets (trigger ratio) and the success rate (hit ratio). However, it is not our intention to optimize any parameters ex post and we use a critical value of 2 throughout the next sections simply following the conventional understanding of extreme.

EXHIBIT 5
Critical Z-Scores and 20-Year Performance R = 2%

Critical t-Value	Trigger Ratio	Hit Ratio	Sharpe Ratio	Annual Excess Return	Information Ratio
3	17.1%	62.2%	0.64	0.14%	0.23
2	61.2%	60.2%	0.70	0.53%	0.66
1.5	89.0%	58.5%	0.70	0.65%	0.75
1	100.0%	58.2%	0.68	0.60%	0.64
0	100.0%	56.3%	0.66	0.57%	0.53

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The Combined Model

The momentum and value strategies are combined in a sequential process shown in Exhibit 6 (the combined model). The sequential model serves as an easy-to-follow example to capture the short-term momentum effects in trending markets (Equation (2)), reduce downside risk by avoiding both negative momentum (Equation (3)) and overvalued asset classes (Equation (5)), and participate in market reversal rallies by investing in extremely undervalued situations (Equation (5)). Throughout our illustrative model, the weights are adjusted from the former step keeping the sum of the asset class weights, including cash, unchanged.[10] All equations are designed to only move exposures between certain asset classes and cash to keep the model simple and avoid overfitting.

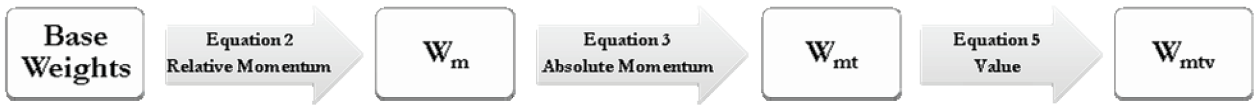
In Equation (5), we further reduce the exposures to overvalued asset classes (Z <-2) to zero from Equation (3) as an effort to reduce downside risk. We increase the corresponding allocation to cash and use it to equally fund all the undervalued asset classes, if there are any. The weights will not be adjusted in this step if there is no over/under-valued asset class.

$$
\begin{aligned}
&W_{mtv}(cash) = W_{mt}(cash) + \sum W_{mt}(j) \\
&W_{mtv}(i) = 0\%, \textit{any asset class } j \textit{ with } z < -2 \\
&n : \textit{number of asset classes with } z > 2 \\
&W_{mtv}(k) = W_{mtv}(k) + \frac{W_{mtv}(cash)}{n}, \textit{asset class } k \textit{ with } z > 2, \textit{if } n > 0 \\
&W_{mtv}(cash) = 0\%, \textit{if } n > 0
\end{aligned}
\tag{5}
$$

The combined model adjusts the importance of the value and the momentum strategies play under different market conditions using the cash reserve. Before an asset class becomes extremely undervalued, it first experiences strong negative momentums. Cash reserve is increased by the modified (absolute) momentum strategy before the nonlinear value strategy is triggered. The combined model also amplifies the role of the value strategy by giving all cash available at the disposal of the value strategy during distressed markets and market bottoms. Exhibit 7 shows the cash levels under the momentum (M), the modified momentum (MT), the value only strategy (V), and the combined model (MT&V) on the equally weighted benchmark. The modified momentum always reserves more cash during highly volatile and distressed markets (for example: 1989 to 1990, 1999, and 2008 to 2009), which were the exact periods during which the nonlinear value strategy is triggered. For example, the relative momentum strategy M reserved about 20% cash in November and December 2008, while the modified momentum strategy MT allocated 100% to cash,[11] which was then in the combined model, fully invested under the nonlinear value strategy. At the same time, the value-only strategy V, strictly following Equation (4), did not fully invest holding about 10% cash.[12] In other words, the combined model (Equation (5)) amplified the nonlinear value strategy during this period by fully investing under it. This simple but effective mechanism to time the importance of the two strategies adds extra value and improves the risk-adjusted performances in the long run.

EXHIBIT 6
Tactical Allocation Models

The Combined Model



The Alternative Model

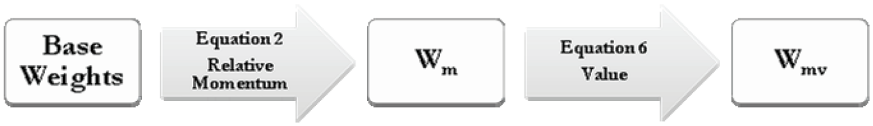


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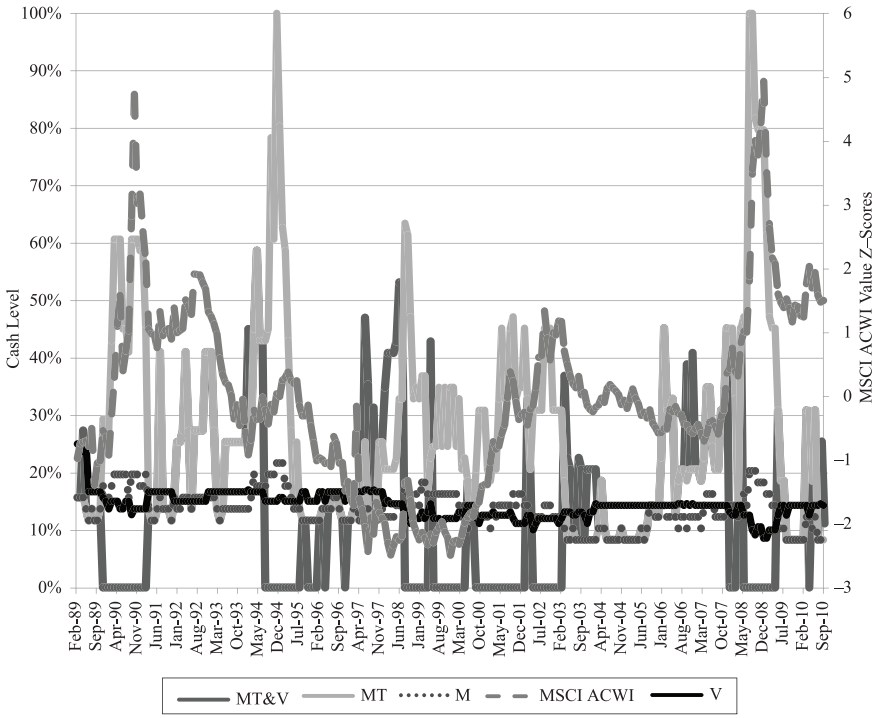
EXHIBIT 7

Historical Cash Levels under Momentum (*M*), Modified Momentum (*MT*), The Value-Only (V), the Combined Model (*MT&V*), and MSCI ACWI Valuation Z-Scores



The Alternative Model

The combined model in the previous section, especially with the modified momentum strategy (Equation (3)), could lead to dramatic allocation changes and large tracking errors, which may not be ideal for some investors. Alternatively, one can choose to combine value and momentum without this step (Exhibit 6: The Alternative Model). The tracking errors are reduced significantly in this alternative model, which also serves as a benchmark to evaluate the timing ability of the combined model.

Starting directly from the relative momentum allocation *Wm*, we only adjust the exposures of over/under-valued asset classes and cash[13]:

$$\begin{gathered} W_{mv}(cash) = W_m(cash) + \sum W_m(j) \\ W_{mv}(j) = 0\%, \quad any\ asset\ class\ j\ with\ z < -2 \\ n: number\ of\ asset\ classes\ with\ z > 2 \\ W_{mv}(k) = Wmv(k) + \frac{W_{mv}(cash)}{n}, \quad asset\ class\ k\ with\ z > 2,\ if\ n > 0 \\ W_{mv}(cash) = 0\%, \quad if\ n > 0 \end{gathered} \tag{6}$$

The 50-50 Model

Finally, we include the equally weighted combination of momentum and value strategies (Equation (7)). The combined weight is simply half of the momentum strategy weight *Wm* and half of the value strategy weight *Wv*. This naive combination spends no effort in timing the importance of the two by keeping them equal at all times. It offers the smallest deviation from the benchmarks since the tactical adjustments from value and momentum tend to offset each other because of the opposite nature of the two strategies, meaning that the asset class with higher (lower) momentum rank usually has lower (higher) valuation rank. Nevertheless, it offers highly consistent outperformance over the benchmarks.

$$W_{50-50} = 0.5 * W_m + 0.5 * W_v \tag{7}$$

MAIN RESULTS

The main results are presented in Exhibits 8 to 16. *E* and *H* stand for the equally weighted and the

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EXHIBIT 8
Annual Returns (as of 12/31/2010)

	E	*E(V)*	*E(M)*	*E(M&V)*	*E(MT)*	*E(MT&V)*	*E(50−50)*
2010	10.63%	12.11%	12.18%	13.53%	9.77%	12.73%	12.15%
2009	22.35%	25.21%	23.58%	28.66%	13.84%	27.56%	24.41%
2008	−21.83%	−19.01%	−17.62%	−18.15%	−0.59%	−3.79%	−18.31%
2007	7.60%	7.66%	8.20%	8.77%	8.30%	8.87%	7.94%
2006	8.47%	8.79%	11.05%	12.80%	11.46%	11.92%	9.91%
2005	8.93%	8.93%	8.29%	8.29%	8.29%	8.29%	8.62%
2004	13.12%	13.28%	14.29%	14.34%	14.24%	14.30%	13.78%
2003	19.11%	18.52%	20.95%	18.23%	19.46%	15.51%	19.73%
2002	6.02%	6.09%	7.64%	7.94%	7.29%	8.59%	6.87%
2001	−2.27%	−0.78%	0.11%	0.51%	3.39%	4.82%	−0.33%
2000	11.53%	12.92%	14.58%	20.35%	14.91%	22.96%	13.75%
1999	9.55%	7.86%	9.85%	5.75%	8.36%	3.46%	8.86%
1998	−2.87%	−2.89%	−0.53%	1.77%	3.78%	7.25%	−1.71%
1997	7.76%	7.98%	7.87%	10.17%	8.32%	10.63%	7.93%
1996	16.31%	16.32%	18.38%	18.66%	18.38%	18.66%	17.34%
1995	16.31%	16.35%	16.15%	16.77%	15.19%	15.76%	16.25%
1994	2.11%	2.10%	1.08%	0.70%	1.28%	1.52%	1.59%
1993	9.46%	9.46%	11.48%	11.48%	13.12%	13.12%	10.46%
1992	6.69%	6.73%	7.84%	7.84%	7.14%	7.14%	7.28%
1991	17.56%	18.93%	15.90%	18.20%	10.27%	15.86%	17.41%
1990	1.37%	−0.32%	3.17%	0.25%	9.39%	1.59%	1.44%
1989	12.65%	12.17%	13.41%	12.60%	13.61%	12.60%	12.79%
Max DD	33.50%	30.12%	28.61%	29.27%	6.77%	13.20%	29.36%
Period	Jun 08 –Mar 09	Jun 08 –Mar 09	Jun 08 –Mar 09	Jun 08 –Mar 09	Jun 08 –Oct 08	Jun 08 –Mar 09	Jun 08 –Mar 09

hypothetical base allocation, respectively. The strategy is noted in the brackets. *V* and *M* stand for the value-only strategy and the relative momentum strategy. *MT* is the modified momentum strategy following Equations (2) and (3). *MT&V* is the combined model and *M&V* is the alternative model. The equally weighted combination of momentum and value strategies is noted as *50–50*. The testing period is from January 1989[14] to December 2010.

The strategies are basic by design, but nonetheless the results are significant. All strategies outperform the benchmarks for the 22-year testing period. The modified momentum *MT* has the smallest maximum drawdown (Exhibits 8 and 9) and improves the risk-adjusted return significantly in the long run because of capital preservation. It has only one year, 2008, with a negative return out of the 22 years. Its biggest outperformance is in highly volatile and stressed markets, such as 1990 (Gulf War); 1998 (LTCM collapse); 2000 to 2002 (tech bubble burst); and 2008 (financial crisis). However, both the relative and the modified momentum strategies do not perform well in the sharp market-reversal periods, such as 1991, 1999, 2003, and 2009-2010 (see Exhibit 10). On the other hand, with the nonlinear value strategy, the combined model significantly reduced the underperformance of the momentum strategies in 1991 and outperformed the benchmark in 2009. It generates significant positive excess returns in 15 of 22 years. In the combined model, cash reserve is often increased by the modified (absolute) momentum strategy during distressed markets because of strong negative momentum across risky asset classes. During distressed periods, the combined model sits on abundant cash and waits patiently until the valuation is cheap enough—the nonlinear value signal is

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EXHIBIT 9
Annual Returns (as of 12/31/2010)

	H	*H(V)*	*H(M)*	*H(M&V)*	*H(MT)*	*H(MT&V)*	*H(50−50)*
2010	11.80%	12.89%	12.94%	13.46%	6.42%	10.57%	12.92%
2009	27.09%	30.05%	28.46%	30.91%	15.26%	31.83%	29.27%
2008	−28.58%	−25.97%	−24.67%	−24.96%	−4.94%	−8.43%	−25.32%
2007	9.08%	9.15%	9.69%	10.10%	9.61%	10.01%	9.43%
2006	13.26%	13.59%	15.92%	16.57%	16.00%	16.32%	14.75%
2005	9.40%	9.40%	8.74%	8.74%	8.74%	8.74%	9.07%
2004	13.55%	13.71%	14.74%	14.75%	14.68%	14.69%	14.23%
2003	24.12%	23.51%	26.10%	24.66%	21.57%	17.26%	24.81%
2002	−3.41%	−3.34%	−1.82%	−1.72%	5.64%	8.33%	−2.58%
2001	−7.20%	−5.76%	−4.88%	−4.72%	3.73%	5.82%	−5.32%
2000	0.65%	1.92%	3.44%	13.49%	6.82%	19.12%	2.68%
1999	14.25%	12.51%	14.55%	5.19%	14.32%	4.18%	13.53%
1998	6.32%	6.33%	8.82%	13.50%	5.95%	12.25%	7.57%
1997	10.16%	10.39%	10.23%	16.10%	10.38%	16.25%	10.31%
1996	13.08%	13.09%	15.10%	15.14%	15.10%	15.14%	14.09%
1995	17.02%	17.07%	16.85%	16.39%	15.78%	14.68%	16.96%
1994	1.86%	1.86%	0.83%	0.73%	1.55%	4.24%	1.35%
1993	14.38%	14.38%	16.48%	16.48%	16.26%	16.26%	15.43%
1992	1.93%	1.97%	3.04%	3.04%	1.81%	1.81%	2.50%
1991	18.00%	19.35%	16.35%	17.56%	7.78%	14.72%	17.85%
1990	−5.64%	−7.33%	−3.80%	−5.16%	4.54%	−3.99%	−5.55%
1989	12.09%	11.61%	12.86%	12.32%	12.97%	12.32%	12.23%
Max DD	39.11%	36.42%	34.16%	34.71%	7.61%	14.37%	34.88%
	Nov 07	Nov 07	Jun 08	Jun 08	Nov 07	Nov 07	Nov 07
Period	–Mar 09	–Mar 09	–Mar 09	–Mar 09	–Oct 08	–Mar 09	–Mar 09

triggered from at least two standard deviation discounts. It further amplifies the role of the value strategy by fully investing the cash under it. The combined model adds extra value by timing the importance of the nonlinear value and the modified momentum strategies correctly and generates more excess return than the simple sum of the two individually.[15]

The biggest underperformance of the combined model is in 1999 and 2003. In 1999, the Z-scores of the equity market were below –2, which suggested that equity was significantly overvalued. Avoiding overvalued assets, the combined model underweighted equity and underperformed in 1999. However, the equity market crashed in the following three years, 2000 to 2002, during which the combined model generated an average annual excess return of more than 14% on the equity-heavy strategic base allocation *H(MT&V)*—the margin of safety at work. In 2003, the nonlinear approach with a critical value of 2 simply failed to recognize a value situation.[16]

In general, both the combined model *MT&V* and the alternative model *M&V* have higher Sharpe ratios (Exhibits 11 and 12) and better performance (Exhibits 13 and 14) than the *50–50*, which ignores the dynamic role of each strategy under different market conditions. The simple average of the value and momentum strategies has the smallest tracking error due to the opposite nature of the two, which could result in a higher information ratio.

A robustness test is done through a capital asset pricing model (CAPM)-type regression analysis following Equation (7).[17] To examine whether each of the strategies and steps have generated alpha, intermediate results from former step(s) are used as benchmarks, which

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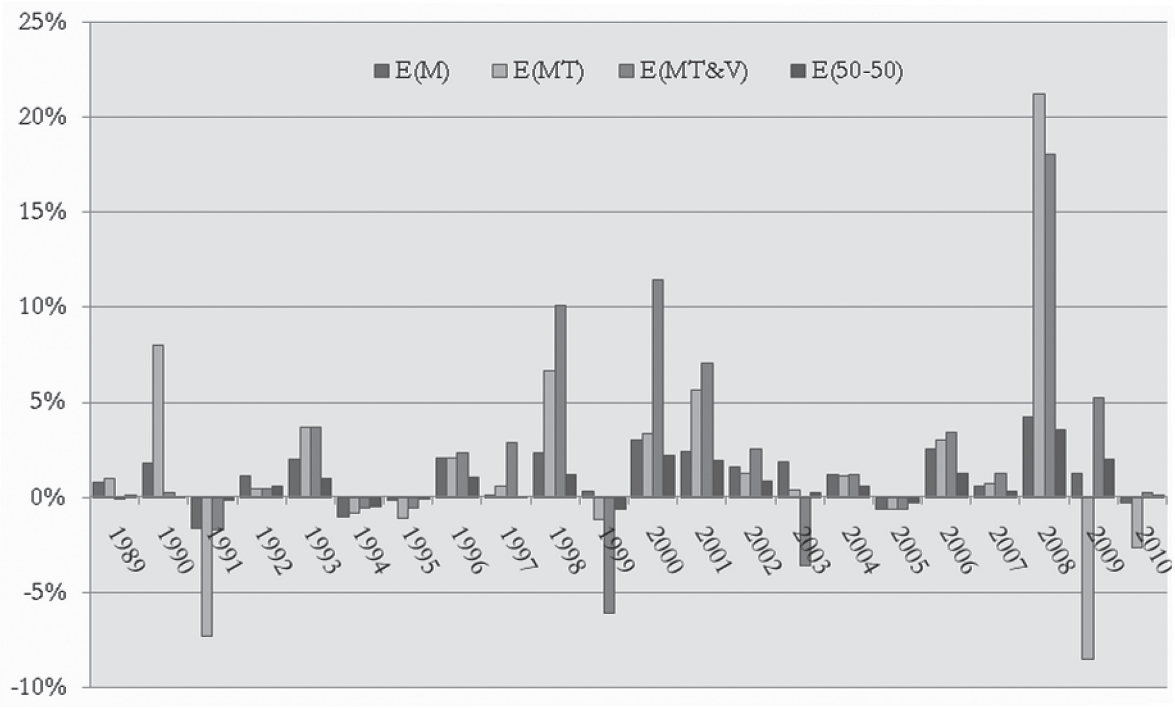
EXHIBIT 10
Annual Excess Returns



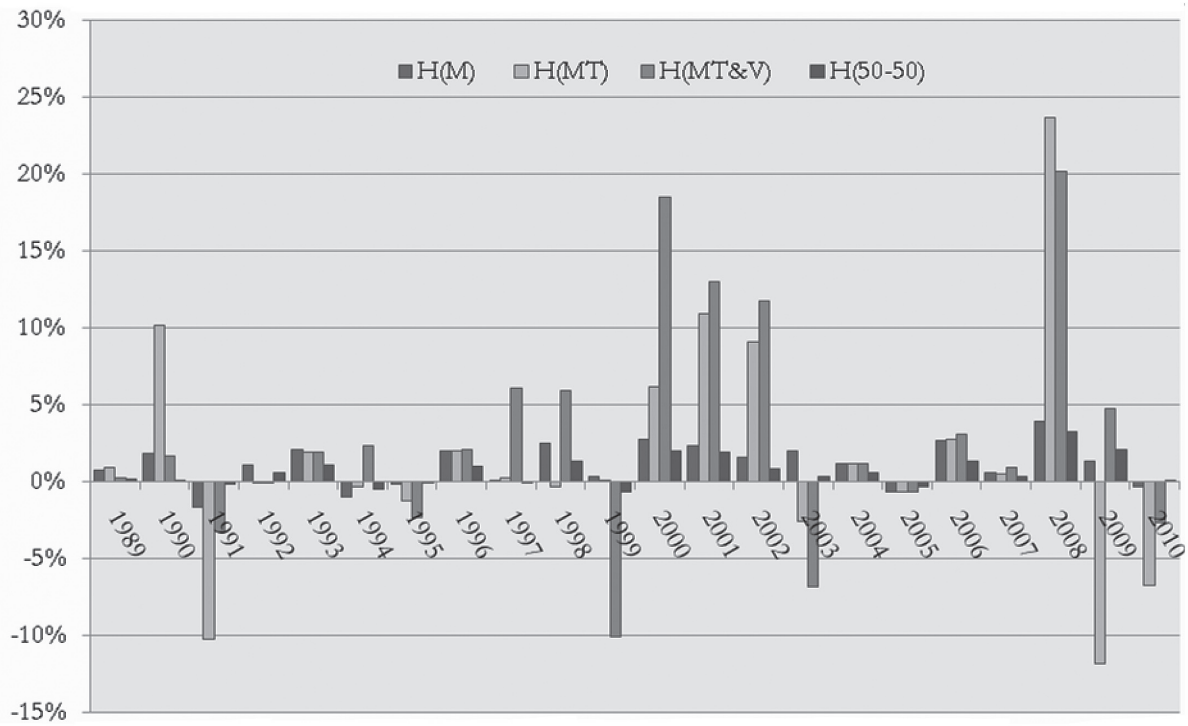


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EXHIBIT 11

Return and Risk R = 2% (as of 12/31/2010)

Years	*E*	*E(V)*	*E(M)*	*E(M&V)*	*E(MT)*	*E(MT&V)*	*E(50-50)*
Returns							
1 yr	10.63%	12.11%	12.18%	9.77%	13.53%	12.73%	12.15%
3 yr	1.90%	4.37%	3.94%	6.90%	5.55%	10.81%	4.17%
5 yr	4.31%	5.90%	6.18%	8.08%	7.60%	10.64%	6.04%
7 yr	6.18%	7.35%	7.60%	8.97%	8.64%	10.81%	7.48%
10 yr	6.50%	7.44%	8.09%	9.23%	8.65%	10.43%	7.77%
15yr	7.09%	7.71%	8.67%	9.69%	9.46%	11.07%	8.19%
20yr	7.87%	8.41%	9.08%	9.59%	9.80%	10.94%	8.75%
2004-2009	3.99%	5.29%	5.79%	8.15%	6.95%	10.12%	5.54%
1999-2004	9.26%	9.80%	11.28%	11.71%	12.04%	13.07%	10.54%
1994-1999	9.18%	8.89%	10.14%	10.68%	10.44%	11.01%	9.51%
1990-1994	7.28%	7.17%	7.76%	8.17%	7.48%	7.69%	7.47%
Volatilities							
1 yr	9.06%	8.88%	9.70%	9.99%	9.00%	9.74%	9.25%
3 yr	14.51%	13.86%	13.06%	14.19%	6.82%	10.14%	13.39%
5 yr	11.65%	11.14%	10.64%	11.46%	6.11%	8.42%	10.83%
7 yr	10.25%	9.82%	9.63%	10.22%	6.19%	7.81%	9.67%
10 yr	9.11%	8.73%	8.55%	9.06%	5.71%	7.20%	8.59%
15yr	8.03%	7.75%	7.66%	7.95%	5.42%	6.55%	7.65%
20yr	7.33%	7.14%	7.02%	7.36%	5.10%	6.38%	7.03%
2004-2009	11.11%	10.60%	10.04%	10.86%	5.32%	7.70%	10.26%
1999-2004	8.74%	8.35%	8.09%	8.61%	5.15%	6.72%	8.16%
1994-1999	5.19%	5.24%	5.26%	4.94%	4.45%	5.50%	5.21%
1990-1994	4.95%	5.22%	4.99%	5.43%	4.89%	5.96%	4.96%
Sharpe Ratio							
1 yr	1.16	1.33	1.24	1.32	1.09	1.28	1.29
3 yr	0.16	0.34	0.36	0.45	1.02	1.07	0.35
5 yr	0.23	0.37	0.45	0.54	1.01	1.03	0.41
7 yr	0.43	0.56	0.62	0.68	1.13	1.12	0.59
10 yr	0.50	0.62	0.73	0.75	1.24	1.15	0.68
15yr	0.51	0.60	0.74	0.81	1.20	1.19	0.67
20yr	0.61	0.70	0.81	0.86	1.18	1.15	0.76
2004-2009	0.33	0.44	0.51	0.58	1.18	1.09	0.48
1999-2004	1.16	1.22	1.46	1.43	1.64	1.48	1.36
1994-1999	0.57	0.53	0.67	0.71	0.89	0.75	0.60
1990-1994	0.52	0.47	0.61	0.51	0.70	0.50	0.55

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EXHIBIT 12

Return and Risk R = 2% (as of 12/31/2010)

Years	*H*	*H(V)*	*H(M)*	*H(M&V)*	*H(MT)*	*H(MT&V)*	*H (50-50)*
Returns							
1 yr	11.80%	12.89%	12.94%	13.46%	6.42%	10.57%	12.92%
3 yr	0.49%	2.81%	3.00%	3.68%	5.25%	10.10%	2.92%
5 yr	4.63%	6.15%	6.80%	7.42%	8.19%	11.30%	6.48%
7 yr	6.53%	7.66%	8.18%	8.63%	9.18%	11.41%	7.92%
10 yr	5.65%	6.55%	7.40%	7.62%	9.43%	11.09%	6.98%
15yr	6.68%	7.28%	8.37%	9.26%	9.77%	11.81%	7.83%
20yr	7.60%	8.12%	8.89%	9.59%	9.44%	11.40%	8.51%
2004-2009	4.17%	5.48%	6.00%	6.51%	8.66%	10.93%	5.75%
1999-2004	4.92%	5.45%	6.93%	8.74%	10.29%	12.93%	6.19%
1994-1999	12.11%	11.82%	13.07%	13.18%	12.24%	12.41%	12.45%
1990-1994	5.75%	5.61%	6.26%	6.15%	6.25%	6.33%	5.94%
Volatilities							
1 yr	12.30%	12.25%	13.16%	13.27%	11.33%	12.25%	12.67%
3 yr	17.26%	16.74%	15.85%	16.49%	8.03%	11.49%	16.24%
5 yr	13.88%	13.47%	12.92%	13.39%	7.36%	9.67%	13.15%
7 yr	12.13%	11.79%	11.51%	11.87%	7.20%	8.92%	11.60%
10 yr	11.27%	10.94%	10.54%	10.82%	6.30%	8.00%	10.70%
15yr	10.20%	9.92%	9.74%	9.41%	6.74%	7.30%	9.79%
20yr	9.41%	9.22%	9.02%	8.80%1	6.48%	7.24%	9.08%
2004-2009	12.96%	12.52%	11.90%	2.38%	6.19%	8.57%	12.16%
1999-2004	10.86%	10.50%	10.01%	10.19%	5.75%	7.29%	10.21%
1994-1999	7.13%	7.04%	7.43%	5.14%	7.07%	6.03%	7.20%
1990-1994	7.71%	8.21%	7.26%	7.59%	5.95%	7.32%	7.66%
Sharpe Ratio							
1 yr	0.97	1.05	1.01	1.03	0.61	0.88	1.03
3 yr	0.08	0.22	0.23	0.27	0.61	0.85	0.22
5 yr	0.24	0.35	0.41	0.44	0.82	0.93	0.38
7 yr	0.41	0.51	0.56	0.58	0.98	1.03	0.54
10 yr	0.36	0.44	0.53	0.54	1.13	1.09	0.49
15yr	0.38	0.45	0.56	0.66	0.96	1.15	0.50
20yr	0.47	0.53	0.62	0.70	0.91	1.07	0.57
2004-2009	0.31	0.41	0.48	0.50	1.14	1.08	0.45
1999-2004	0.45	0.48	0.70	0.73	1.43	1.43	0.59
1994-1999	0.74	0.72	0.79	1.02	0.76	0.91	0.76
1990-1994	0.16	0.14	0.24	0.22	0.28	0.24	0.19

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EXHIBIT 13

Excess Return and Tracking Error R = 2% (as of 12/31/2010)

Years	*E(V)*	*E(M)*	*E(M&V)*	*E(MT)*	*E(MT&V)*	*E(50−50)*
Excess Return						
1 yr	1.49%	1.55%	2.91%	−0.86%	2.10%	1.52%
3 yr	2.48%	2.63%	4.24%	5.60%	9.53%	2.56%
5 yr	1.59%	2.23%	3.66%	4.13%	6.70%	1.91%
7 yr	1.17%	1.68%	2.72%	3.05%	4.90%	1.43%
10 yr	0.94%	1.77%	2.32%	2.91%	4.11%	1.36%
15 yr	0.62%	1.70%	2.50%	2.73%	4.11%	1.16%
20 yr	0.53%	1.30%	2.01%	1.81%	3.16%	0.92%
2004–2009	1.30%	1.80%	2.96%	4.16%	6.13%	1.56%
1999–2004	0.54%	2.02%	2.78%	2.45%	3.81%	1.28%
1994–1999	−0.29%	0.96%	1.26%	1.51%	1.84%	0.34%
1990–1994	−0.11%	0.48%	0.20%	0.89%	0.41%	0.19%
Tracking Error						
1 yr	1.32%	2.31%	2.43%	3.79%	3.69%	1.47%
3 yr	1.48%	2.88%	2.35%	12.50%	8.31%	1.69%
5 yr	1.19%	2.42%	2.17%	9.82%	6.70%	1.42%
7 yr	1.02%	2.19%	2.00%	8.34%	5.75%	1.32%
10 yr	0.97%	2.09%	2.21%	7.33%	5.63%	1.23%
15 yr	0.88%	1.85%	2.31%	6.16%	5.06%	1.06%
20 yr	0.81%	1.66%	2.09%	5.49%	4.66%	0.95%
2004–2009	1.05%	2.34%	2.08%	9.70%	6.58%	1.42%
1999–2004	0.73%	1.65%	2.53%	3.40%	4.55%	1.01%
1994–1999	0.62%	1.08%	2.19%	2.53%	3.99%	0.52%
1990–1994	1.13%	1.41%	1.13%	3.81%	3.25%	0.44%
Information Ratio						
1 yr	1.12	0.67	1.20	−0.23	0.57	1.04
3 yr	1.67	0.92	1.81	0.45	1.15	1.52
5 yr	1.34	0.92	1.69	0.42	1.00	1.35
7 yr	1.15	0.77	1.36	0.37	0.85	1.08
10 yr	0.97	0.84	1.05	0.40	0.73	1.10
15 yr	0.70	0.92	1.08	0.44	0.81	1.10
20 yr	0.66	0.78	0.96	0.33	0.68	0.97
2004–2009	1.25	0.77	1.42	0.43	0.93	1.10
1999–2004	0.73	1.23	1.10	0.72	0.84	1.27
1994–1999	−0.46	0.89	0.58	0.60	0.46	0.65
1990–1994	−0.09	0.34	0.18	0.23	0.13	0.44

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EXHIBIT 14

Excess Return and Tracking Error R = 2% (as of 12/31/2010)

Years	*H(V)*	*H(M)*	*H(M&V)*	*H(MT)*	*H(MT&V)*	*H(50−50)*
Excess Return						
1 yr	1.09%	1.14%	1.65%	−5.38%	−1.23%	1.12%
3 yr	2.32%	2.51%	3.19%	4.76%	9.61%	2.42%
5 yr	1.52%	2.17%	2.79%	3.56%	6.67%	1.85%
7 yr	1.12%	1.65%	2.09%	2.64%	4.87%	1.39%
10 yr	0.90%	1.75%	1.96%	3.78%	5.44%	1.33%
15yr	0.60%	1.69%	2.57%	3.09%	5.13%	1.15%
20yr	0.52%	1.29%	1.99%	1.84%	3.80%	0.91%
2004−2009	1.31%	1.82%	2.34%	4.49%	6.76%	1.57%
1999−2004	0.53%	2.01%	3.82%	5.37%	8.01%	1.27%
1994−1999	−0.28%	0.96%	1.08%	0.14%	0.31%	0.34%
1990−1994	−0.13%	0.51%	0.41%	0.51%	0.58%	0.20%
Tracking Error						
1 yr	0.82%	1.96%	1.96%	6.25%	5.38%	1.22%
3 yr	1.36%	2.79%	2.27%	14.56%	9.88%	1.75%
5 yr	1.10%	2.35%	2.01%	11.33%	7.79%	1.45%
7 yr	0.94%	2.13%	1.87%	9.61%	6.66%	1.30%
10 yr	0.91%	2.06%	1.95%	9.28%	7.71%	1.27%
15yr	0.84%	1.82%	3.73%	7.74%	7.42%	1.09%
20yr	0.78%	1.64%	3.27%	6.91%	6.68%	0.97%
2004−2009	1.05%	2.34%	2.00%	10.98%	7.44%	1.42%
1999−2004	0.73%	1.65%	3.75%	6.79%	8.49%	1.01%
1994−1999	0.62%	1.08%	5.02%	2.28%	6.42%	0.52%
1990−1994	1.13%	1.41%	1.05%	5.98%	4.47%	0.44%
Information Ratio						
1 yr	1.32	0.58	0.84	−0.86	−0.23	0.92
3 yr	1.70	0.90	1.40	0.33	0.97	1.38
5 yr	1.39	0.92	1.39	0.31	0.86	1.27
7 yr	1.19	0.77	1.12	0.27	0.73	1.07
10 yr	0.99	0.85	1.01	0.41	0.71	1.05
15yr	0.71	0.93	0.69	0.40	0.69	1.05
20yr	0.67	0.79	0.61	0.27	0.57	0.94
2004−2009	1.25	0.78	1.17	0.41	0.91	1.11
1999−2004	0.72	1.22	1.02	0.79	0.94	1.26
1994−1999	−0.46	0.89	0.21	0.06	0.05	0.66
1990−1994	−0.12	0.36	0.39	0.08	0.13	0.44

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EXHIBIT 15
Robustness Test

Strategy	Benchmark	Alpha	T-Stat*	Beta	T-Stat	Hit Ratio
E(V)	*E*	0.49%	1.96	0.97	131.69	60.25%
E(M)	*E*	1.73%	3.98	0.93	67.71	61.60%
E(50-50)	*E*	1.11%	4.61	0.95	127.42	62.74%
E(M&V)	*E*	1.95%	3.95	0.97	56.13	62.36%
E(MT)	*E*	6.11%	4.53	0.47	13.96	60.46%
E(MT&V)	*E*	5.24%	4.46	0.69	20.34	59.70%
H(V)	*H*	0.37%	1.91	0.99	173.27	60.25%
H(M)	*H*	1.68%	3.98	0.94	91.60	61.60%
H(50-50)	*H*	1.03%	4.41	0.96	170.85	62.74%
H(M&V)	*H*	2.63%	3.19	0.88	45.44	59.32%
H(MT)	*H*	6.05%	3.56	0.46	14.28	61.22%
H(MT&V)	*H*	6.69%	4.28	0.55	16.73	58.94%
E(MT)	*E(M)*	4.43%	3.88	0.59	20.90	54.22%
E(MT&V)	*E(MT)*	1.47%	1.81	0.93	19.34	54.76%
H(MT)	*H(M)*	4.73%	3.04	0.55	18.65	54.22%
H(MT&V)	*H(MT)*	3.00%	2.06	0.82	17.54	54.76%

**For our sample size, the critical t-value is about 1.65.*

means using simple relative momentum *M* as benchmark for the modified momentum strategy *MT*, and using *MT* as benchmark for the combined model *MT&V*.

$$R_p - R_f = \alpha + \beta(R_b - R_f) \tag{7}$$

All strategies generate significant alpha with high hit ratios on both benchmarks (Exhibit 15). Momentum strategy by nature adjusts weights every month, while value strategy was triggered about 60% of the time under the nonlinear approach with a critical value of 2. The *50–50* combination of the two has the highest hit ratio, which is visible in the consistent outperformance in 17 of the 22 years with an average underperformance of only 35 bps (basis points) in the other 5 years.

Exhibit 16 shows the growth of one dollar under all the strategies. Performance is quite stable over time, staying above benchmarks through various market cycles. By including the value strategy in a nonlinear fashion, the combined model works better than the value and momentum strategies individually. It identifies market bottoms and improves performance during sharp market reversal periods by increasing the role of the value strategy.

The adjusting unit *R* serves as the control for risk and leverage. It decides how much the tactical allocation deviates away from the base allocation (Equations (2) and (4)). Exhibit 17 shows the allocation of the combined model (including cash) on the equally weighted base *E(MT&V)* through the entire test period with $R = 2\%$. As *R* increases to a certain value, the no-short and no-leverage constraints need to be relaxed.[18] The risk associated with making bigger bets and more volatile tactical allocation also increases correspondingly. This effect can be demonstrated by plotting the 20-year Sharpe ratio against *R* (Exhibit 18). The Sharpe ratio is maximized around $R = 5\%$ at a value of 1.16.

CONCLUSION AND DISCUSSION

Our model provides dynamic top-down insights into tactical asset allocation. The basic yield valuation measurement to each asset class is standardized using the Z-score. Such standardization not only enables us to directly compare valuation measures across asset classes, but offers insight about each asset class's absolute valuation by its own standard. Together with the nonlinear approach, it helps to identify market peaks and bottoms for each asset class. We improve the momentum strategy by considering both relative and absolute performances. In the combined tactical asset allocation model, this modification adds value by adjusting the importance of value and momentum strategies under different market conditions. We also provide alternative models for achieving smaller tracking errors.

However, investors must take further consideration and care on these instructive models before implementation, in light of issues such as liquidity constraints, transaction costs, and taxes. Several improvements are possible. The holding period of the value strategy can be optimized with some mean-reversion process modeling. Other indicators of market risk (Wang, Sullivan, and Ge [2012]; and Sullivan, Peterson, and Waltenbaugh [2010]) can be used for better market timing than using absolute performances. For now, static covariance structures are implied in the base allocations, which means the equally weighted and the more conventional allocation. We can further consider a dynamic and conditional covariance structure. The Black-Litternman framework can be used when combining the views from the value and momentum strategies by relating the value Z-scores to the confidence levels.

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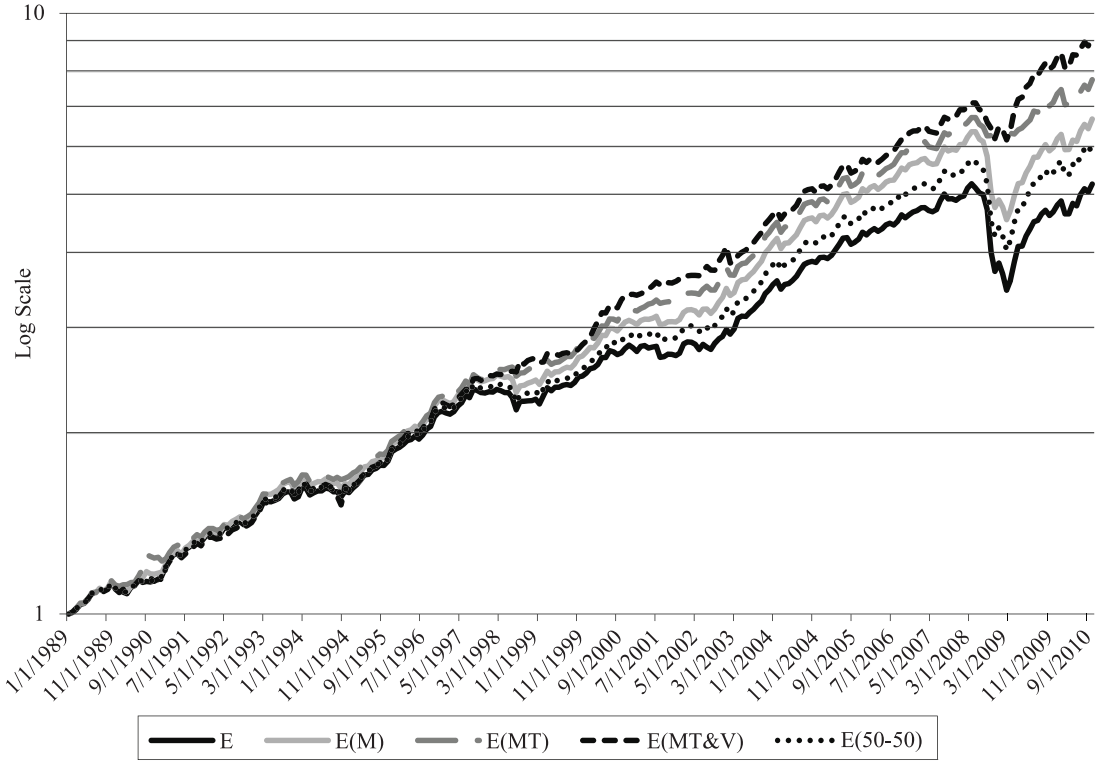
EXHIBIT 16
Growth of One Dollar



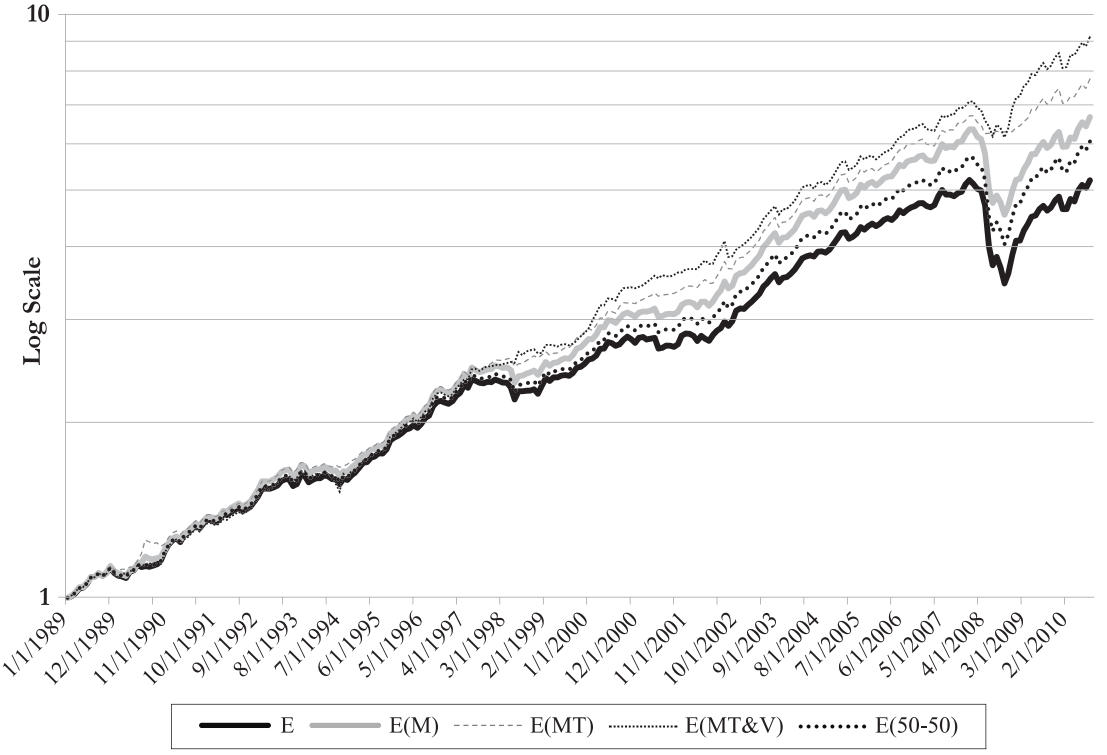


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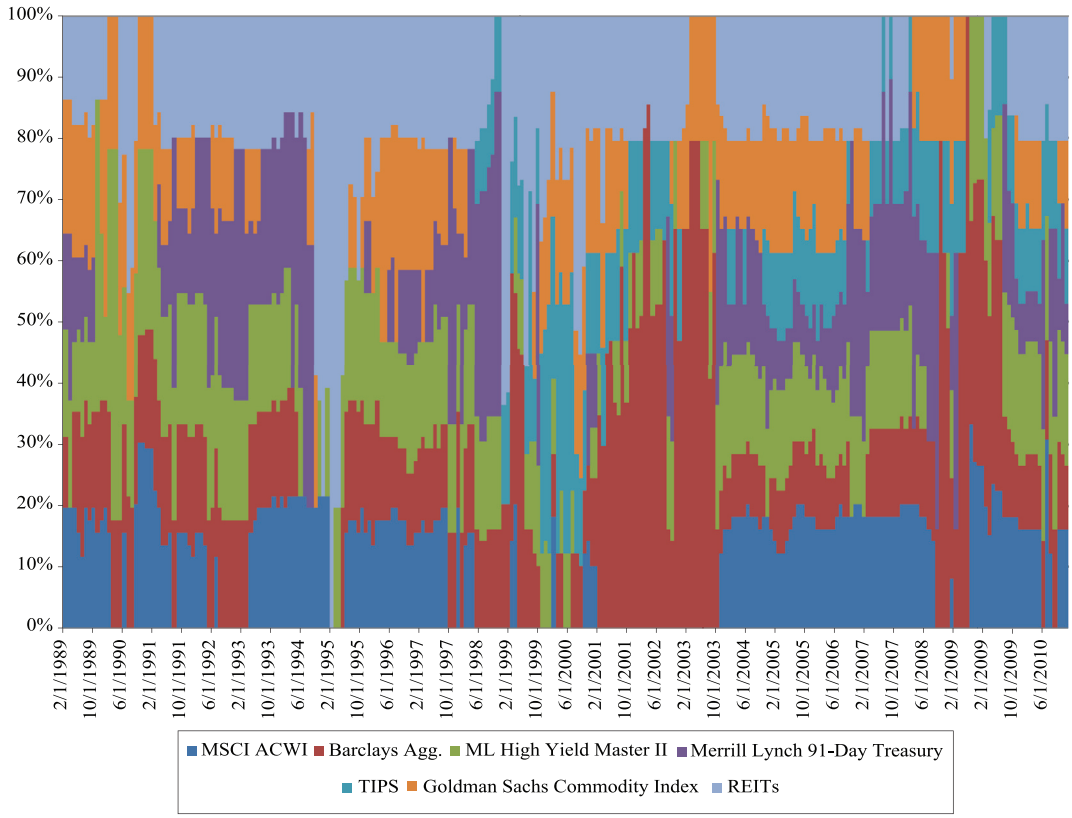
EXHIBIT 17

Weight Evolution of *E (MT&V) R* = 2%



For a color version of this exhibit, please visit The Journal of Wealth Management website at www.iijournals.com/jwm.

EXHIBIT 18

Sharpe Ratio vs. *R*

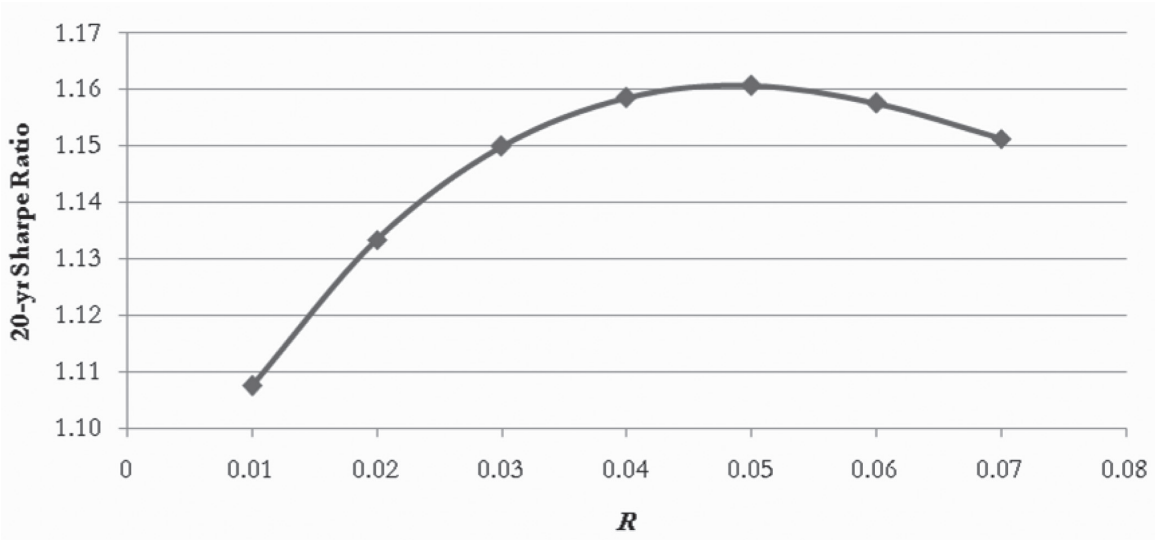


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EXHIBIT 19
ETFs

Asset Class	ETF	Mgr/Issue	Exp. bps
Global Equity	ACWI	BlackRock	35
Investment Grade	AGG	BlackRock	24
High Yield	JNK*	SSgA	40
Cash	–	–	–
TIPS	TIP	BlackRock	20
Commodity	GSP	Barclays	75
Real Estate	VNQ	Vanguard	13

**JNK tracks the price and yield performance of the Barclays Capital High Yield Very Liquid Index.*

EXHIBIT 20
Annual Return Using ETFs R = 2% (as of 12/31/2010)

	1 Year	2 Year		1 Year	2 Year
E	11.10%	15.09%	***H***	11.88%	17.60%
E(M&V)	12.64%	17.43%	***H(M&V)***	12.11%	18.66%
E(MT&V)	12.08%	16.10%	***H(MT&V)***	9.83%	16.98%

EXHIBIT 21
Tactical Allocation from *E(MT&V) R* = 2%

	Global Equity	Investment Grade	High Yield	Cash	TIPs	Commodity	Real Estate
12/1/2010	16.29%	10.29%	18.29%	8.29%	12.29%	14.29%	20.29%
11/1/2010	16.29%	12.29%	18.29%	22.57%	0.00%	10.29%	20.29%
10/1/2010	16.29%	14.29%	18.29%	8.29%	12.29%	10.29%	20.29%
9/1/2010	0.00%	16.29%	18.29%	30.86%	14.29%	0.00%	20.29%
8/1/2010	12.29%	16.29%	18.29%	18.57%	14.29%	0.00%	20.29%
7/1/2010	30.86%	16.29%	20.29%	0.00%	18.29%	0.00%	14.29%
6/1/2010	0.00%	14.29%	18.29%	30.86%	16.29%	0.00%	20.29%
5/1/2010	16.29%	10.29%	18.29%	8.29%	12.29%	14.29%	20.29%
4/1/2010	16.29%	12.29%	18.29%	8.29%	10.29%	14.29%	20.29%
3/1/2010	16.29%	12.29%	18.29%	8.29%	10.29%	14.29%	20.29%
2/1/2010	16.29%	12.29%	18.29%	8.29%	14.29%	10.29%	20.29%
1/1/2010	16.29%	10.29%	18.29%	8.29%	12.29%	14.29%	20.29%
12/1/2009	16.29%	10.29%	18.29%	8.29%	12.29%	14.29%	20.29%
11/1/2009	18.29%	10.29%	20.29%	8.29%	12.29%	14.29%	16.29%
10/1/2009	18.29%	12.29%	20.29%	18.57%	14.29%	0.00%	16.29%
9/1/2009	18.29%	14.29%	20.29%	18.57%	12.29%	0.00%	16.29%
8/1/2009	18.29%	16.29%	20.29%	30.86%	14.29%	0.00%	0.00%
7/1/2009	22.57%	40.86%	20.29%	0.00%	16.29%	0.00%	0.00%
6/1/2009	22.57%	40.86%	20.29%	0.00%	16.29%	0.00%	0.00%
5/1/2009	23.57%	43.86%	14.29%	0.00%	18.29%	0.00%	0.00%
4/1/2009	15.36%	35.64%	15.36%	0.00%	18.29%	0.00%	15.36%
3/1/2009	19.93%	40.21%	19.93%	0.00%	0.00%	0.00%	19.93%
2/1/2009	26.57%	46.86%	26.57%	0.00%	0.00%	0.00%	0.00%
1/1/2009	26.57%	46.86%	26.57%	0.00%	0.00%	0.00%	0.00%
12/1/2008	27.24%	45.52%	27.24%	0.00%	0.00%	0.00%	0.00%
11/1/2008	33.33%	33.33%	33.33%	0.00%	0.00%	0.00%	0.00%
10/1/2008	0.00%	100.00%	0.00%	0.00%	0.00%	0.00%	0.00%
9/1/2008	0.00%	61.43%	0.00%	0.00%	18.29%	20.29%	0.00%
8/1/2008	0.00%	61.43%	0.00%	0.00%	18.29%	20.29%	0.00%
7/1/2008	0.00%	16.29%	0.00%	45.14%	18.29%	20.29%	0.00%
6/1/2008	8.29%	16.29%	14.29%	12.29%	18.29%	20.29%	10.29%
5/1/2008	0.00%	49.14%	12.29%	0.00%	18.29%	20.29%	0.00%
4/1/2008	0.00%	61.43%	0.00%	0.00%	18.29%	20.29%	0.00%
3/1/2008	0.00%	61.43%	0.00%	0.00%	18.29%	20.29%	0.00%
2/1/2008	0.00%	16.29%	0.00%	45.14%	18.29%	20.29%	0.00%
1/1/2008	14.29%	16.29%	0.00%	30.86%	18.29%	20.29%	0.00%

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APPENDIX

IMPLEMENTATION THROUGH EXCHANGE-TRADED FUNDS (ETFs)

Through indexed ETFs (Exhibit 19), our model offers a low cost and easily accessible way for potentially better performance without the complications of hedge fund and private-equity-type managers (Rittereiser and Kochard [2010]).

Using index data as inputs for the model, the returns with real ETFs are shown in Exhibit 20. Since the inception of iShares ACWI is April 2008, the performances using ETFs only have about two-year history. The most recent allocation of the combined model on the equally weighted benchmark [*E(MT&V)*] are shown in Exhibit 20. Despite the highly difficult two-year period in which most tactical allocation models experienced strong market reversal, the combined model, especially on the equally weighted base allocation, still delivered strong performance with well-diversified and dynamic allocations using real ETF returns.[19]

ENDNOTES

The authors would like to thank Rodney Sullivan, Michael Barry, Cliff Asness, Mebane Faber, Bobby Pornrojnangkool, Nick Gerow, and the members of the Investment Office at Georgetown University for their valuable comments.

[1]One can also use 10-month, 6-month, 3-month, and so on. For simplicity, 12-month is used as an example.

[2]Valuation data for emerging market started in January 1986.

[3]The use of NAV-to-price data based on UBS respective of U.S. REIT universe will not affect the conclusions.

[4]Blitz and Van Vliet add/subtract adjustment factors to/from these value measures, which could introduce a forward-looking bias to some level.

[5]A higher Z-score means a better valuation.

[6]Ensure 12 months of return data for the momentum strategy for TIPS.

[7]Also leads to larger tracking errors.

[8]Except when applied to the hypothetical policy weights, momentum caused small negative exposure to TIPS during 2005 to 2006.

[9]Or any desired lower bound.

[10]The sum is 100% for our two strategic allocation portfolios. However, it could be any number from the investor's choice of strategic allocation.

[11]All asset classes except cash experienced negative recent performances. The valuation of cash is not relevant in the combined model since the cash reserve level is exclusive decided by Equations (3) and (5).

[12]In this case, the cash level is decided by its rank of its valuation Z-score just as other asset classes are in Equation 4.

[13]In the alternative model, we still avoid the exposures to overvalued asset classes, since the nonlinear valuation-based tactical change is not as frequent.

[14]Total return data of MSCI ACWI is from December 1988.

[15]The value-only strategy usually has a higher allocation to cash and does not fully invest during the market bottoms, which can be seen in the previous section discussing the combined model.

[16]One can always use a smaller critical value, for example 1.5, to pick up more undervalued situations. However, again, it is not our intention to optimize any parameters post ante.

[17]The monthly risk-free rate is downloaded from Fama–French website.

[18]Under the no-short constraint, the maximum value allowed for R is about 4.76% for the equal-weighted benchmark.

[19]The performance difference between the ETF and the index could be due to the tracking error, being traded at premium/discount to NAV, and the management fees.

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